

Pathmika Weerarathna

Associate Tech Lead — AI/ML Engineer

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Dear Hiring Manager,

I am writing to express my strong interest in AI/ML engineering opportunities with your organization. I am an AI/ML engineer and Associate Tech Lead at Omobio, having self-initiated and led applied AI/ML capabilities in telecom fraud detection from the ground up — progressing from individual learning to leading an AI/ML R&D team. I bring hands-on experience building production NLP/embedding-based detection systems, GenAI subsystems on local LLMs, and ensemble ML classifiers, backed by nearly five years of full-stack software engineering fundamentals.

In my current role, I lead the AI/ML R&D team enhancing our rule-based SMS Firewall product with machine-learning-driven detection, and serve as technical subject-matter expert for AI/ML pre-sales engagements across Africa, South Asia, and the Middle East. As Senior Software Engineer, I built a production SMS fraud-detection module — live for Orange/Moov Côte d'Ivoire — combining SentenceTransformer embeddings and cosine-similarity matching to auto-block OTP-bypass and grey-route traffic, sustaining ~378 messages/second.

Key highlights of my career include:

- **Production ML Systems** — Extended the fraud-detection engine with a local-LLM GenAI subsystem (Llama 3 8B / Google Gemma via Ollama/LangChain) and an XGBoost/CatBoost ensemble classifier with SHAP explainability and Optuna hyperparameter tuning.
- **Applied Fraud Detection** — Contributed AI/ML risk-scoring (Random Forest classification, social-graph SIM-box-ring detection) to an R&D flash-call fraud-detection system built on CAMEL/CAP signaling.
- **Predictive Analytics** — Built a working churn/CLV-prediction prototype (scikit-learn Random Forest, 80–90% validation accuracy) with a Flask API and React dashboard, supporting a 16-module predictive-analytics platform proposal.
- **Engineering Leadership** — Mentor a cross-functional team of 10+ developers on ML engineering practices, code review, and CI/CD, while maintaining full-stack fundamentals across React Native and Next.js/React.

My technical foundation is backed by a B.Sc. in Information Systems (Second Class Upper Division) from the University of Colombo School of Computing, and I am pursuing an M.Sc. in Artificial Intelligence at the University of Moratuwa, applying that knowledge directly in my day-to-day work.

My entry into AI/ML with no prior background, and progression to team leadership, shows I approach unfamiliar domains as opportunities. I would welcome the opportunity to discuss how my background aligns with your team's goals. Thank you for considering my application.

Kind regards,

Pathmika Weerarathna

Associate Tech Lead, Omobio (Pvt) Ltd

B.Sc. Information Systems — University of Colombo School of Computing | M.Sc. Artificial Intelligence — University of Moratuwa